

AlphaZero for Protein Engineering

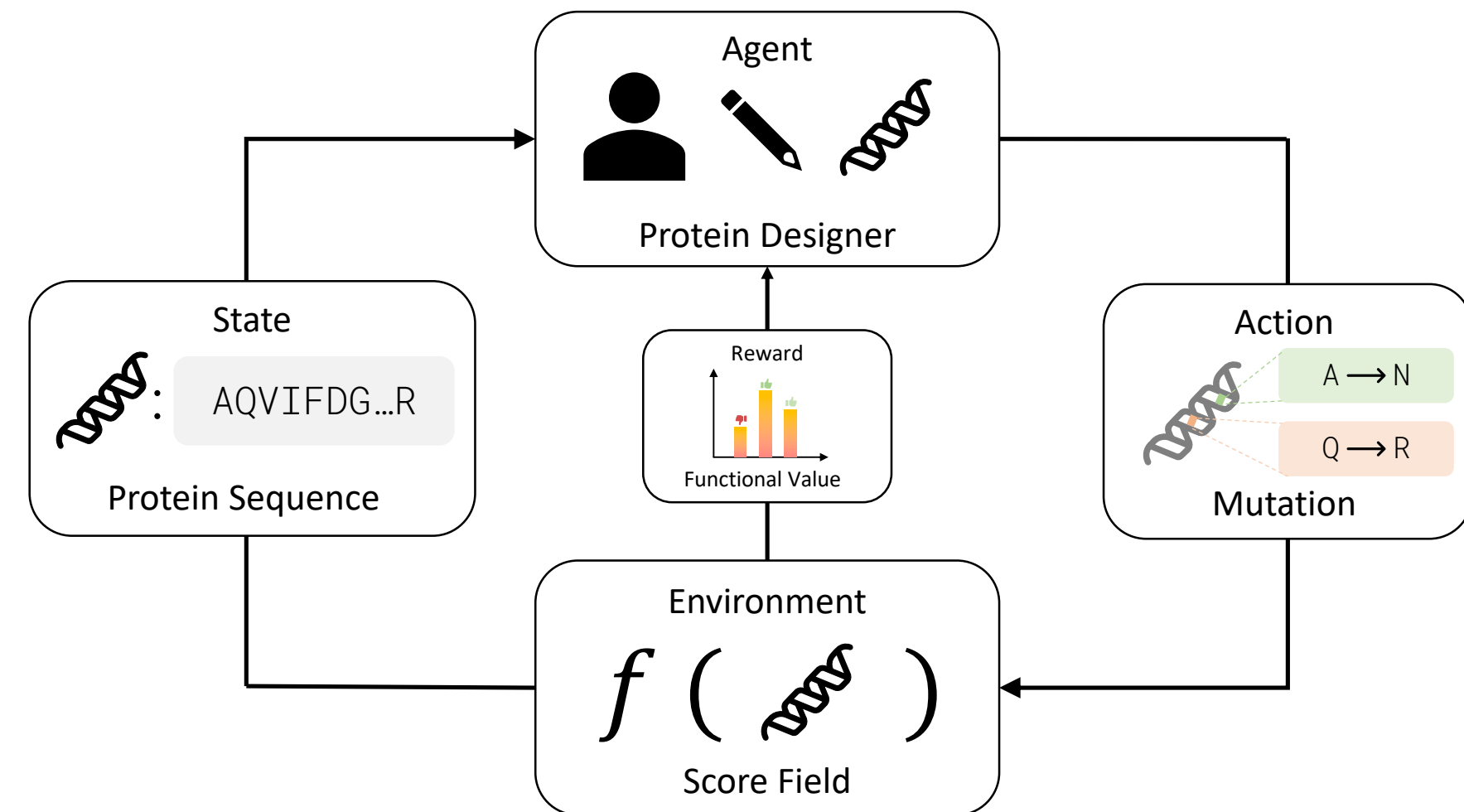
Yifan Qin*, Yan Zeng*, Haizhao Dai*



上海科技大学
ShanghaiTech University

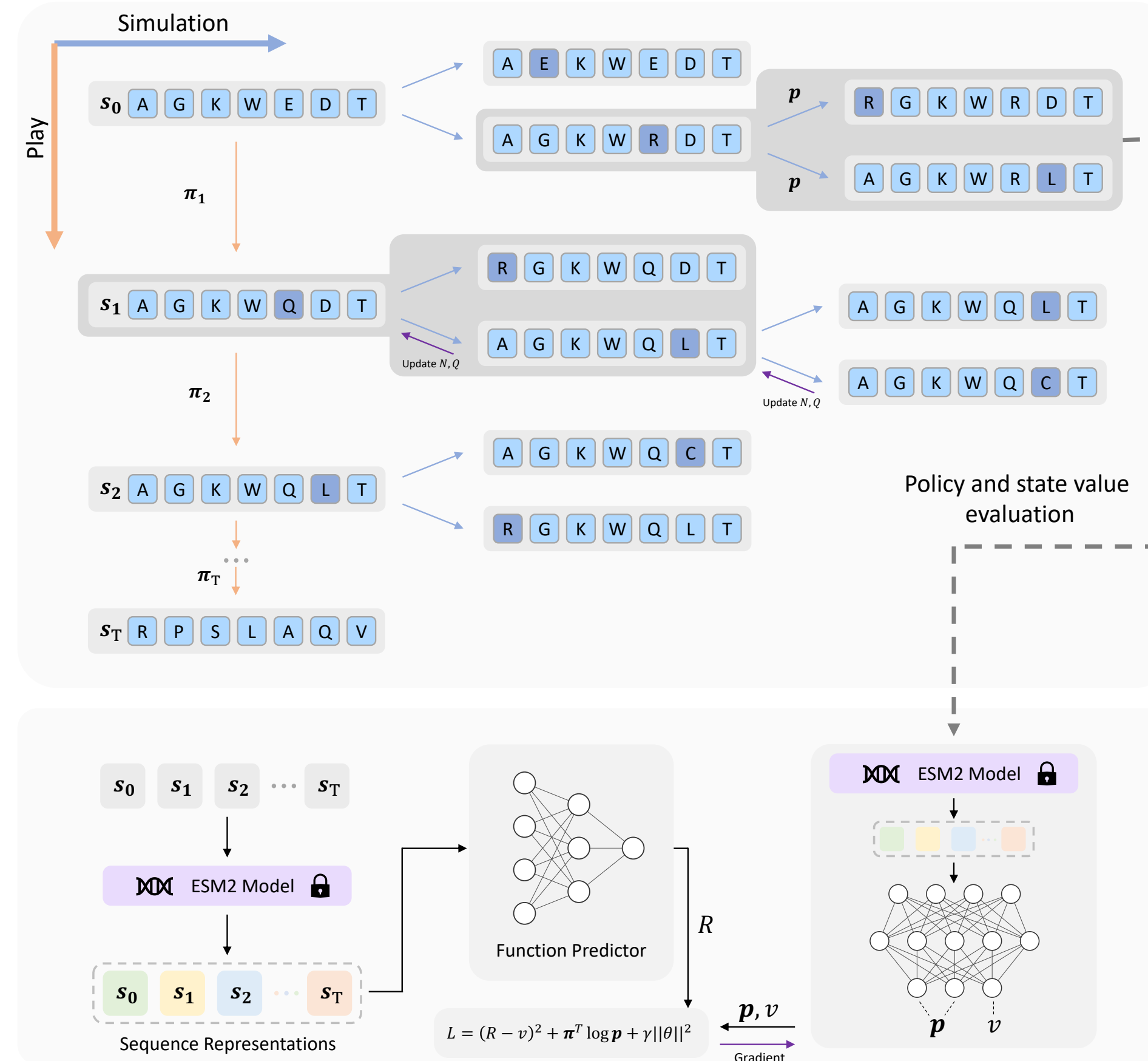
Abstract & Overview

Protein engineering confronts the challenge of navigating vast sequence spaces to discover functional proteins. While reinforcement learning (RL) offers a promising framework where an agent optimizes sequences through mutations against structural/functional rewards, existing approaches like EvoPlay suffer from limited feature extraction due to their reliance on 1D convolutional neural networks (CNNs), which fail to capture long-range residue dependencies or leverage semantic representations. To overcome this, we enhance EvoPlay by integrating ESM2, a Transformer-based protein language model pretrained via masked language modeling. Our framework replaces 1D CNNs with ESM2's self-attention mechanism to extract semantically rich, structure-aware embeddings; redesigns the agent and environment using ESM2's multimodal features (mapped via linear layers to action-space probabilities and rewards); and demonstrates superior in silico performance, generating sequences with higher functional scores than the original EvoPlay. This work advances computational protein design by unifying ESM2's representation power with RL-driven optimization.



- **Agent:** an algorithmic model that learns a sequence optimization strategy.
- **State:** the specific protein sequence being evaluated or modified.
- **Action:** an action that specifies a change in the type of amino acid at a particular position in the sequence. In the field of protein engineering, it is called mutation.
- **Environment:** a computational module that receives new sequences generated by an agent, predicts its 3D structure using a structure prediction tool, evaluates the structure's properties by applying a predefined scoring function, and ultimately feeds the computed scoring value to the agent as a Reward.
- **Reward:** the output value of the scoring function, reflecting the degree to which the current sequence satisfies the design goals.

Pipeline



- Node selection:

$$U_{s,a} \leftarrow c \cdot P_{s,a} \frac{\sqrt{N_{s,a}^{\text{Parent}}}}{1 + N_{s,a}}, Q_{s,a} \leftarrow \frac{Q_{s,a} N_{s,a} + v}{N_{s,a} + 1}$$

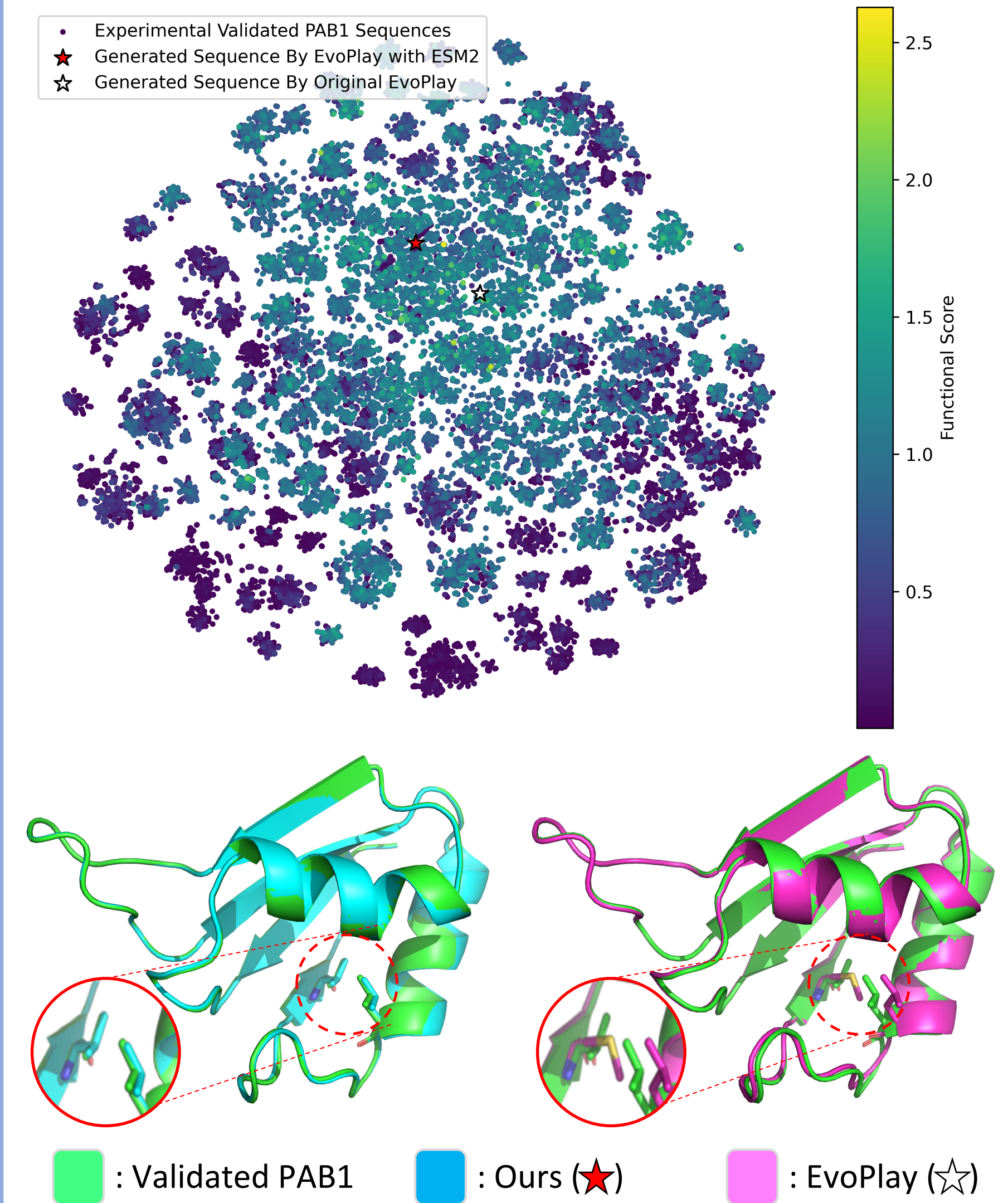
- c is a constant controlling the balance between exploration and exploitation.
- $P_{s,a}$ is the prior probability of action a in state s .
- $N_{s,a}$ is the visit count.
- v is the value of the state which is the output of the policy-value network.

- Loss function:

$$\mathcal{L}(\theta) = \sum_t [(r_t - v_{s_t})^2 - \pi_t^T \log P_{s_t, a}] + \gamma ||\theta||^2$$

- r_t is the reward which is given by the surrogate mode.
- π_t is the selected action probability
- θ is the network parameters.

Results



Designed sequence	score
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYVAP	1.1789
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDEHGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYVAP	1.0942
GNIFIKNLHPDIDNKALYMTSTFSGDILSSKIATDKHGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYVAP	0.8755
GNIFIKNLHPDIDNKALYDTSTFSGDILSSKIATDKHGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQTIYVAP	0.7909
GNIFIKNLHPDIDNKALYDTSTFSGDILSSKIATDKHGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYVAP	0.7744
GNIFIKNLHPDIDNKALYDTSTFSGDILSSKIATDKHGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYVAP	0.7628
GNIFIKNLHPDIDNKALYDTSTFSGDILSSKIATDKHGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYVAP	0.7351
GNIFIKNLHPDIDNKALYDTSTFSGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYCAP	0.7211
GNIFIKNLHPDIDNKALYDTSTFSGDILSSKIATDKHGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYCAP	0.7167
GNIFIKNLHPDIDNKALYDTSTFSGDILSSKIATDKHGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQNIYVAP	0.7134

Top functional score produced by ours. (mean: 0.8366)

Designed sequence	score
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYVAP	1.1446
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYVAP	1.0727
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYMAP	0.9037
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYMAP	0.8616
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYVAP	0.7029
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYVAP	0.6990
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYVAP	0.6778
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYVAP	0.6570
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYCAP	0.5099
GNIFIKNLHPDIDNKALYDTFSVFGDILSSKIATDENGKSKGFGFVHFEFEAAKEAIDALKGMLLNGQEIYMAP	0.4843

Top functional score produced by EvoPlay. (mean: 0.7714)